Blue Crab (*Callinectes sapidus*)

Summary Scores

- **Sensitivity:** Moderate
- **Exposure:** Low
- **Vulnerability:** Low

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.12	2.7	
Complexity in Reproductive Strategy	3.36	2.66	
Dispersal of Early Life Stages	1.6	2.9	
Egg and Larval Survival and Settlement Requirements	2.72	2.9	
Historical Variability in Temperature	1.8	2.7	
Juvenile Habitat Requirements	2.2	2.96	
Other Stressors	2.84	2.65	
Population Growth Rate	1.28	2.59	
Prey Specificity	1.08	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	2.48	2.7	
Spawning Cycle	2.08	2.8	
Stock Size/Status	1.88	1.4	
Total Sensitivity | Certainty	**Moderate (2) | 0.79**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.28	0.90	
Bottom Oxygen	2	0.76	
Bottom Salinity	2.88	0.74	
Bottom Aragonite Solubility	1.1	0.28	
Net primary production	1.12	0.19	
Downward Flux of Particulate Organic Carbon	1.23	0.33	
Total Exposure	**Low (1.48)**	**0.73**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Low (2.97)**	**1.47**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Blue Crab. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	High	Moderate	Decrease
Exposure	Very High	Low	Decrease
Vulnerability	Very High	Low	Decrease
Directional Effect	Neutral	Neutral	None

Stocks

Blue Crab is managed as a single stock or stock information was not considered for this iteration of the CVA.

Score Descriptions:

Sensitivity:

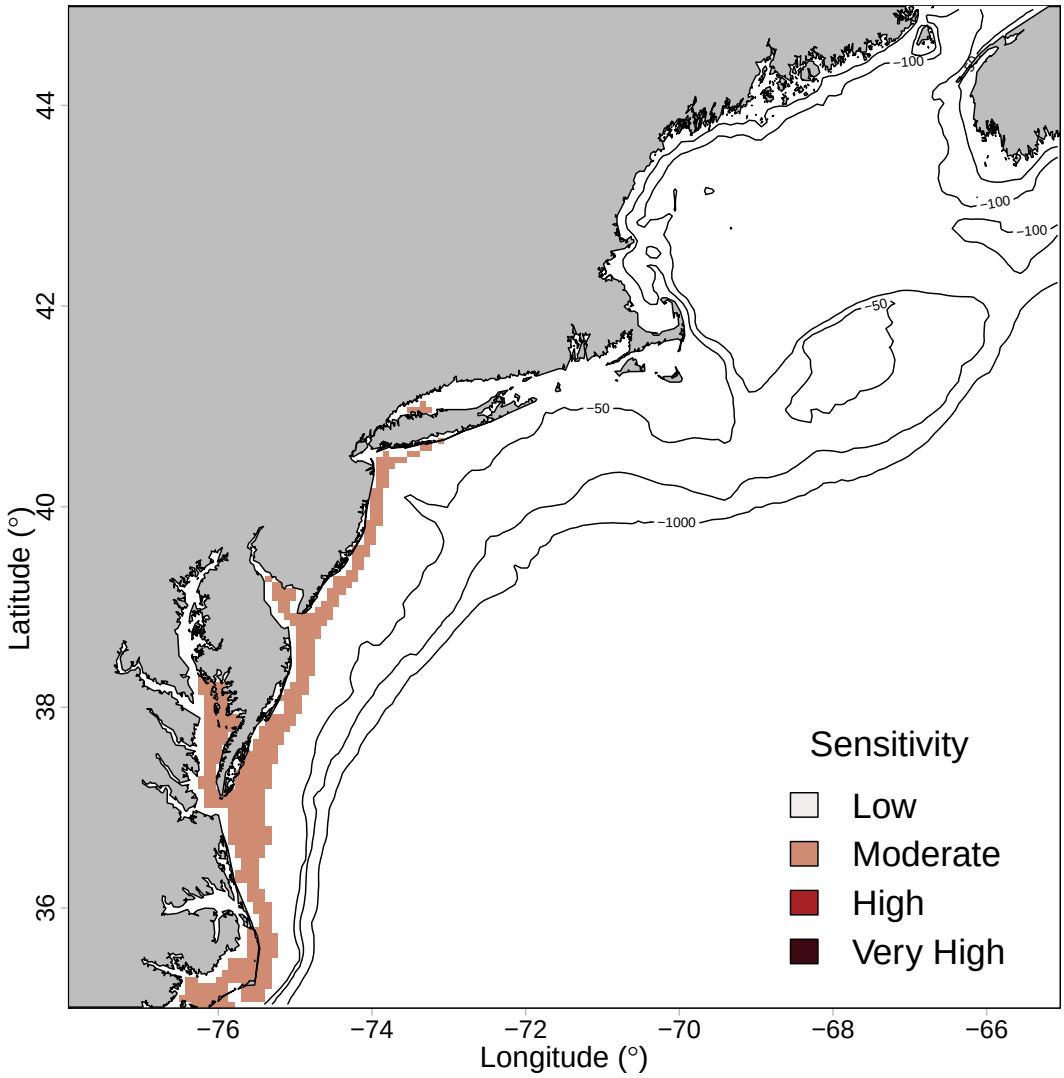


Figure 2: Blue Crab Sensitivity on the Northeast Shelf

Blue crabs had a **Moderate** sensitivity score, with 79% certainty. Three attributes received scores greater than 2.5: Complexity in Reproductive Strategy (3.4), Other Stressors (2.8), and Egg and Larval Survival and Settlement Requirements (2.7) (Figure 1).

Blue crabs rely on estuaries throughout their life cycles, including for spawning. Estuaries are highly vulnerable habitats due to urbanization impacts such as hypoxia, loss of seagrass beds, and changes to freshwater inputs (Guyant et al. 2022). Temperatures have been shown to impact spawning behaviors and larvae require low salinities (10-30) for proper development (Hill, Fowler, and Van Den Avyle 1989).

Blue crabs received a High sensitivity score in the previous Climate Vulnerability Assessment. This was driven by similar attributes to our scores: Other Stressors and Complexity in Reproductive Strategy (Hare et al. 2016).

Exposure:

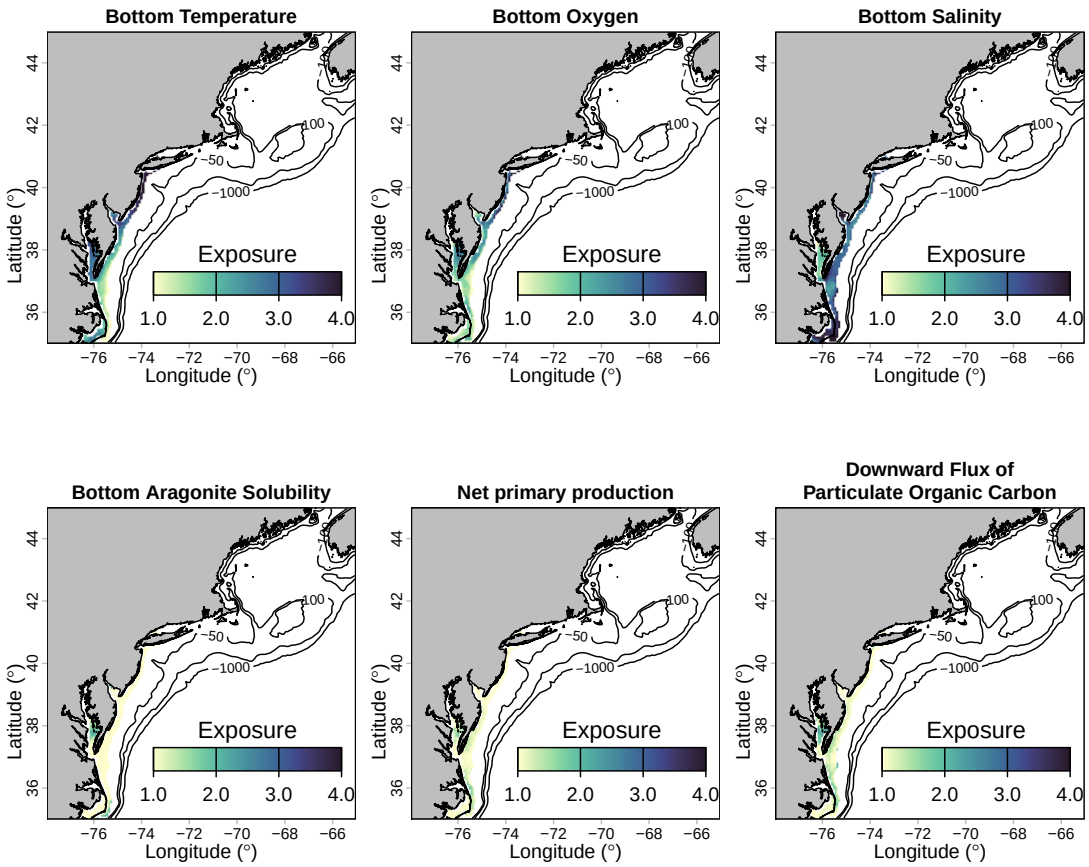


Figure 3: Maps of each Exposure Factor for Blue Crab. Exposure was calculated between 2014-2023 and 2025-2035

Mean Exposure score for the Blue crab was Low (1.48 +/- 0.73) (Figure 1). Exposures were greatest off the coast of New Jersey and within Chesapeake Bay, but Low offshore (Figure 4).

In the previous Assessment, blue crab received a Very High score due to high sea surface temperature, air temperature, and ocean acidification exposure scores (Hare et al. 2016).

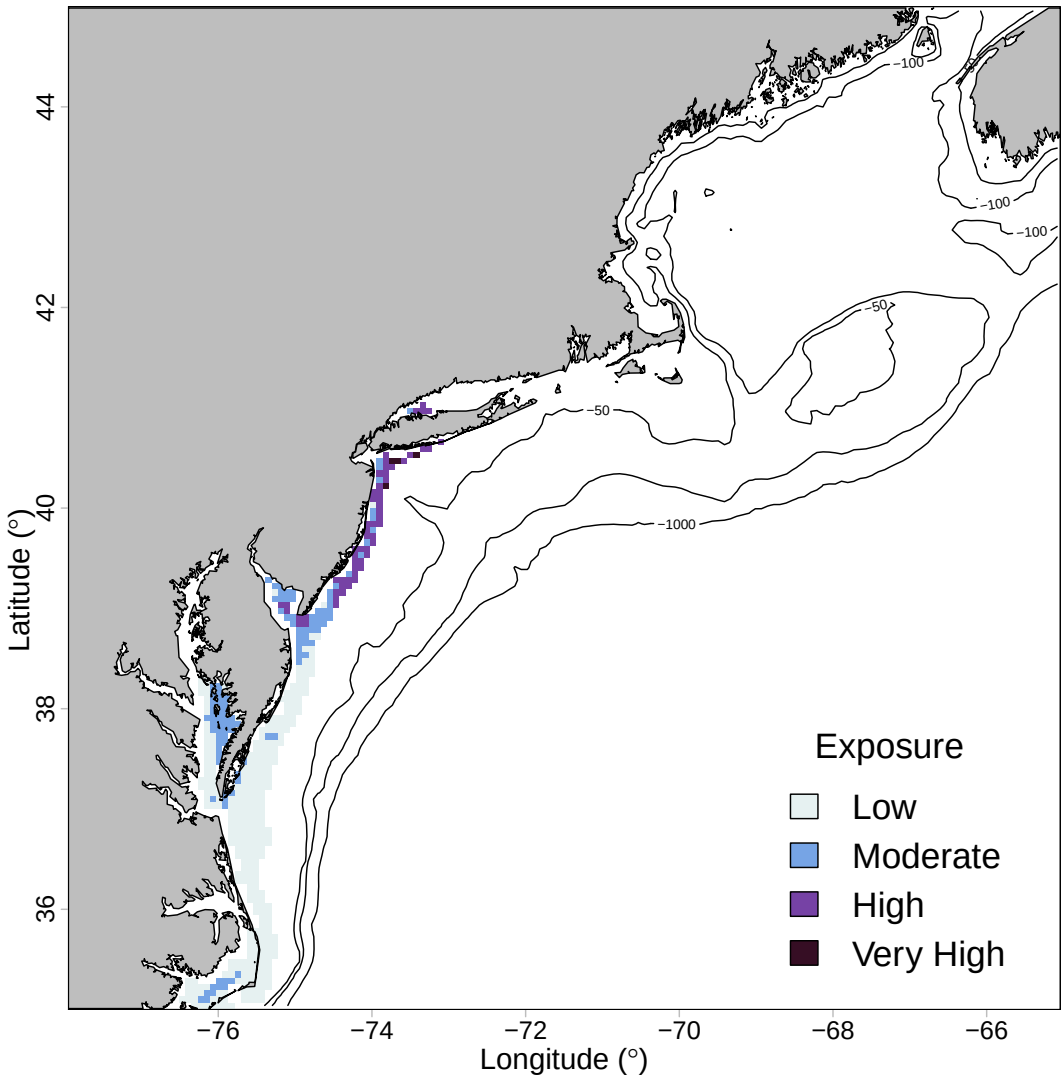


Figure 4: Total Exposure on the Northeast Shelf for Blue Crab. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

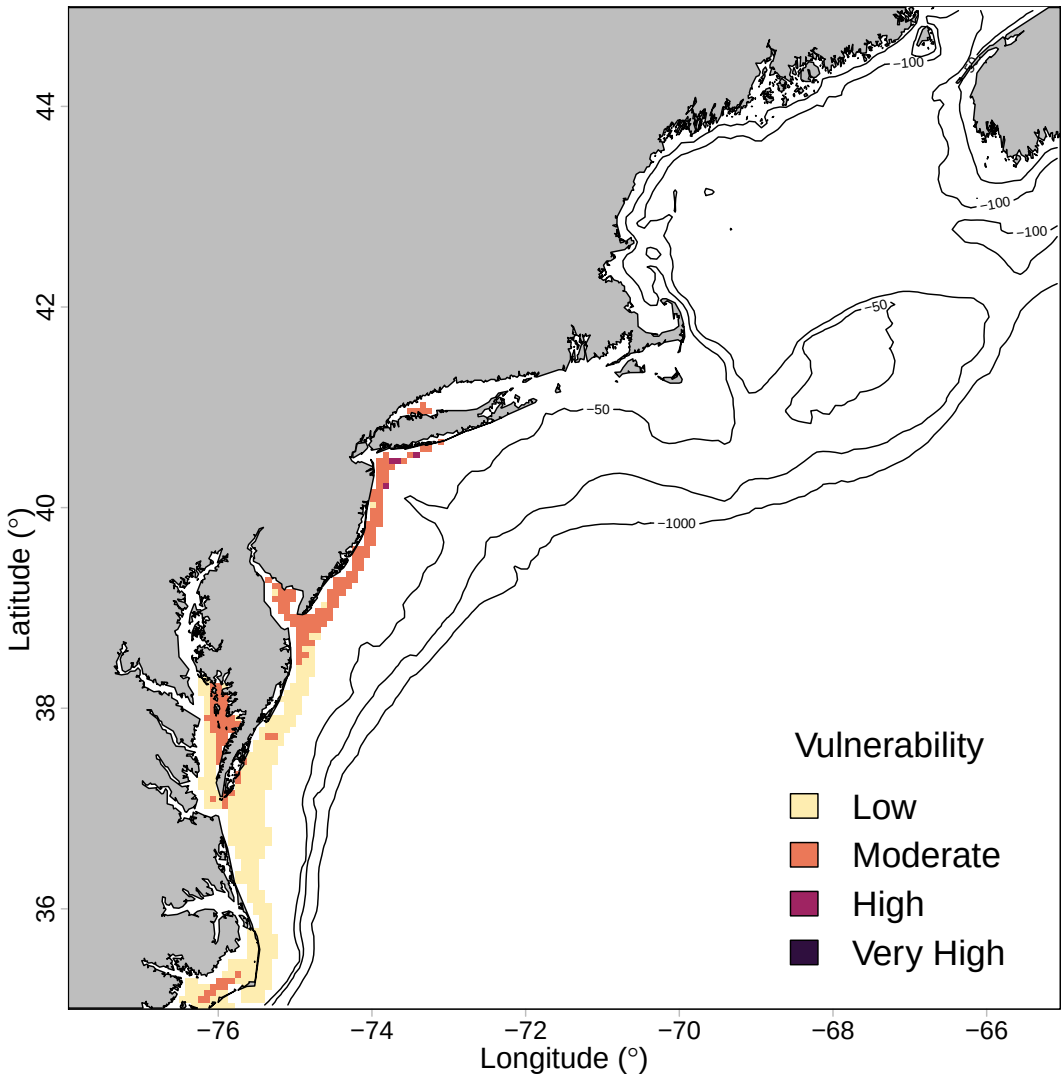


Figure 5: Vulnerability on the Northeast Shelf for Blue Crab. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean Vulnerability for Blue crab was **Low** (2.97 +/- 1.47) due to low Exposure scores (Figure 1). Vulnerability remained Low when Exposure scores were calculated with only important factors (2.61 +/- 1.18) (Figure 1).

In the previous Assessment, Blue crab received a Very High Vulnerability score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

93% of data quality scores were greater than or equal to 2, indicating excellent data quality. Only Stock Size/Status was considered a data gap (mean attribute data quality score < 2) (Figure 1).

Model Confidence Score:

The blue crab model received a 2.6 (+/- 0.2) mean model confidence score, indicating high confidence in the model. Experts said the model generally aligned with expectations for the species distribution on this spatial scale. The ensemble model AUC score was 0.98 indicating excellent model performance.

Additional Metrics

Distribution Shifts:

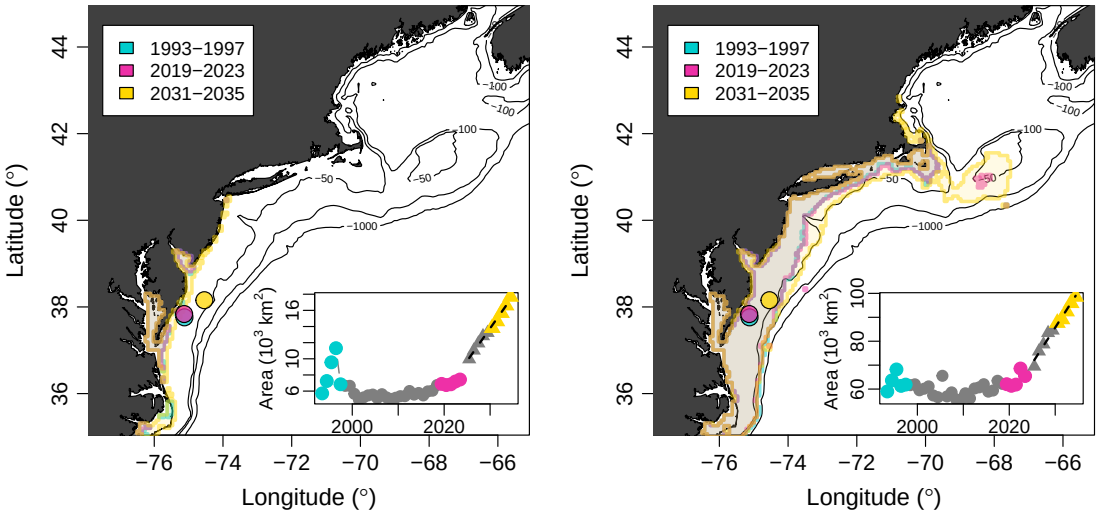


Figure 6: Distribution shifts on the Northeast Shelf for Blue Crab. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Blue crab distributions were stable between 1993 - 2023, with small changes to overall area and range (Figure 6). The weighted centroid shifted 8.5 km north.

Blue crab distributions are expected to shift significantly under near future climate conditions. Significant increases in available habitat are predicted. Blue crab distributions are predicted to expand north and east. Core habitat is predicted to expand north along the coast of New Jersey while the species range is predicted to expand as far east as Georges Bank and as far north as Cape Cod Bay (Figure 6). As a result, the weighted centroid is predicted to shift nearly 63 km east-northeast.

Blue crab received a High Distributional Vulnerability Rank in the previous Assessment (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	8,190	15,723 (+8,680)	5.42	4.01
2019-2023	8.5 km N	7,043 (-1,146)	64,251 (+487)	5.61 (+0.18)	4.33 (+0.32)
2031-2035	62.7 km ENE	15,723 (+8,680)	93,435 (+29,184)	5.92 (+0.31)	6.57 (+2.24)

Directional Effect:

Blue crab are expected to be **neutrally** impacted by climate change. Expert scores had 93% certainty. This is consistent with the previous Assessment (Hare et al. 2016). While degradation of estuarine habitats may limit food sources or force crabs into new areas, warmer temperatures are predicted to increase survival rates, particularly over winter (Aguilar et al. 2008; Glandon, Kilbourne, and Miller 2019).

Climate Effects on Abundance and Distribution

As an estuary dependent species, blue crab habitats have been degraded through urbanization impacts including hypoxia, reduction in seagrass beds, and changes in timing and volume of freshwater inputs. Low oxygen conditions may force crabs into shallower, more spatially restricted areas of Chesapeake Bay, increasing their vulnerability to predation or capture. Prolonged hypoxia may limit food resources (mainly infaunal bivalves) in affected areas even after normoxic conditions resume (Aguilar et al. 2008). Glandon, Kilbourne, and Miller (2019) estimated water temperatures in Chesapeake Bay through 2100 and found that a 50% shorter winter will increase overwinter survival rates in blue crab by 20%. Faster growth, a longer growing season, and lower winter mortality would lead to increased productivity (Glandon, Kilbourne, and Miller 2019). Blue crabs may also adapt phenologically to warming temperatures by shifting the timing of biological events such as spawning (Schneider, Fabrizio, and Lipcius 2024). Blue crabs have been observed in the Gulf of Maine, indicating a possible range extension from climate-driven rising temperatures (Johnson 2015).

Life History Synopsis

The blue crab's native range is along the western Atlantic coast from Nova Scotia to Argentina, including the Gulf of America (Hill, Fowler, and Van Den Avyle 1989). The blue crab fishery is the most valuable fishery in the Chesapeake Bay (Hines 2007). In Chesapeake Bay, adults spawn from May to October, however spawning is likely more limited in cooler waters (University of Maryland Center for Environmental Science (UMCES) 2011). To spawn, adults aggregate in low salinity areas of upper estuaries to mate. Adult females then travel to higher salinity spawning grounds (Hill, Fowler, and Van Den Avyle 1989). The timing of the migration varies with water temperature and occurs earlier in the year at lower latitudes (Aguilar et al. 2008). Eggs (7-14 days) and larvae (40-70 days) are pelagic (Hill, Fowler, and Van Den Avyle 1989) and are carried by currents offshore before using physical transport methods to return to bays and eventually estuaries (Millikin and Williams 1984). Larvae require temperatures of 20-30°C

and salinities of 10-30 ppt for proper development (Hill, Fowler, and Van Den Avyle 1989). Settlement into estuarine habitats is associated with Ekman flow from southward wind events resulting in strong subtidal flow into the estuaries. Once in the estuary, settlement and metamorphosis of the megalopae are stimulated by chemical cues associated with appropriate nursery habitat (Epifanio and Cohen 2016). Larvae are phytoplanktivorous, consuming dinoflagellates and copepod nauplii (Hill, Fowler, and Van Den Avyle 1989). They are preyed upon by a large number of estuarine and marine animals , including other commercially important species such as the striped bass (Hill, Fowler, and Van Den Avyle 1989). Once reaching the juvenile stage, blue crabs gradually migrate into shallower, less saline waters in upper estuaries and rivers, where they grow and mature. Males, which prefer low-salinity waters, generally migrate farther upstream than do females, which tend to stay in the lower rivers and estuaries (Hill, Fowler, and Van Den Avyle 1989). Post-larval and juvenile crabs are considered general scavengers, bottom carnivores, detritivores, and omnivores, consuming a wide range of dead and live fish, crabs, organic debris, shrimp, mollusks (including mussels, clams, oysters, and snails), and aquatic plants (Hill, Fowler, and Van Den Avyle 1989). Adult blue crab prey is very similar to juveniles; they will consume most available prey and freely switch among preferred prey types (Hill, Fowler, and Van Den Avyle 1989). Juvenile and adult blue crabs are consumed by mammals, sea turtles, birds, and several fishes (Hill, Fowler, and Van Den Avyle 1989). Blue crabs reach maturity at 12-18 months, with a maximum lifespan around 4 years of age and a maximum size of 9 inches carapace length (CL) (University of Maryland Center for Environmental Science (UMCES) 2011) (Millikin and Williams 1984). Adults prefer temperatures of 9-10°C for feeding, growth, and migration (Hines 2007). In the Northwest Atlantic, blue crabs are managed by the Maryland Department of Natural Resources, Virginia Marine Resources Commission, and Potomac River Fisheries Commission. These entities work together with NOAA to conduct stock assessments for blue crab. The Chesapeake Bay blue crab population is not overfished based on current biological reference points, but abundance and recruitment remain low. A benchmark stock assessment is underway to improve understanding of blue crab population dynamics and its fishery (Chesapeake Bay Stock Assessment Committee (CBSAC) 2025).

Supplemental Scores

Exposure Timeseries

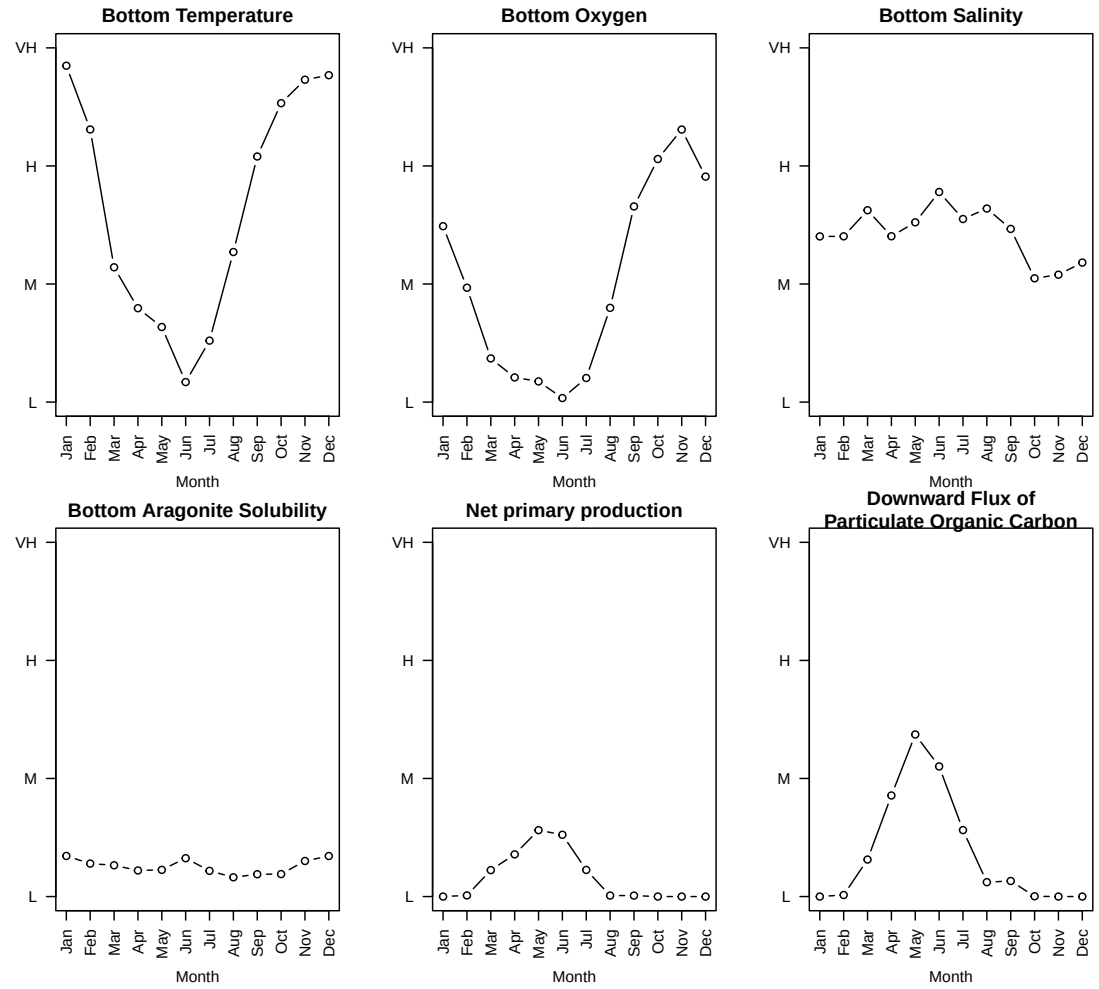


Figure 7: Monthly Timeseries of each Exposure Factor for Blue Crab. Exposure was calculated between 2014-2023 and 2025-2035

Exposure was Low most months but ranged between Moderate to High from September through December (Figure 8). This was due to increases in bottom temperature and oxygen exposure during this time (Figure 7).

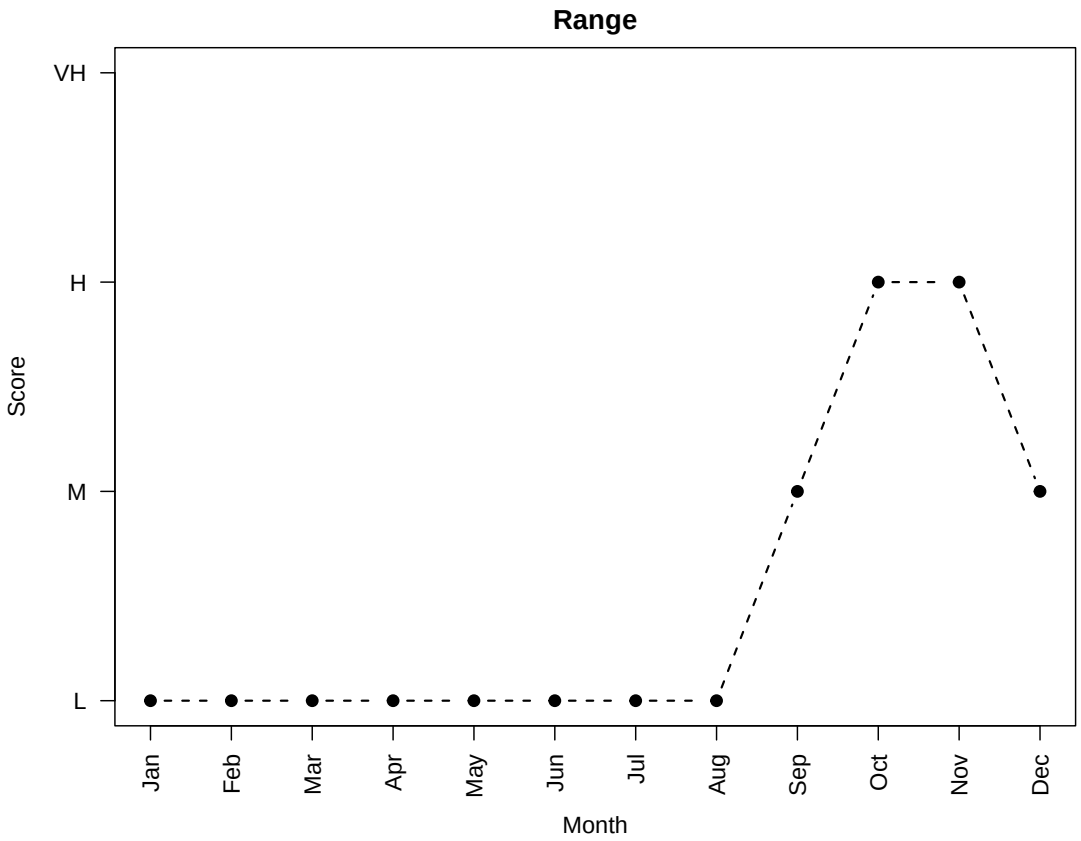


Figure 8: Timeseries of Total Exposure for Blue Crab

Exposure due to Significant Factors

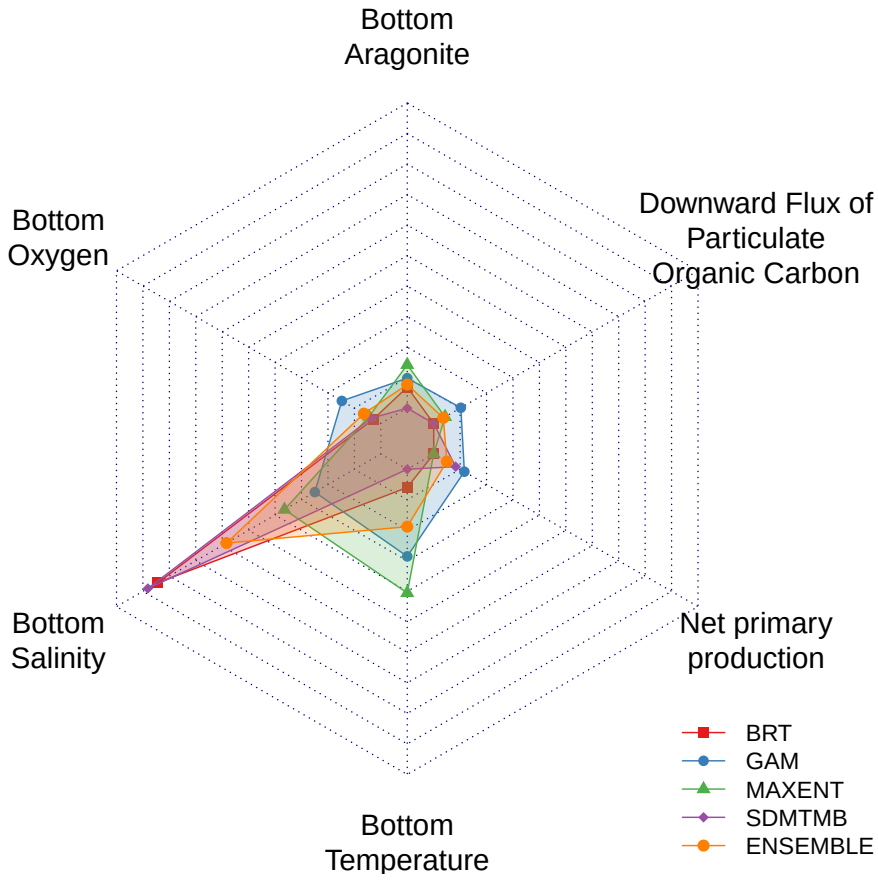


Figure 9: Dynamic Factors driving Blue Crab distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature and salinity were the major dynamic drivers of species distributions (Figure 9). As an estuarine species, blue crab are sensitive to changes in salinity and temperatures play a role in spawning behavior and development (Hill, Fowler, and Van Den Avyle 1989; Aguilar et al. 2008). Both bottom temperature and salinity experienced relatively high exposure along the coast of New Jersey (Figure 10, Figure 3). However, when exposure was calculated with only these factors, overall exposure remained Low (Figure 11).

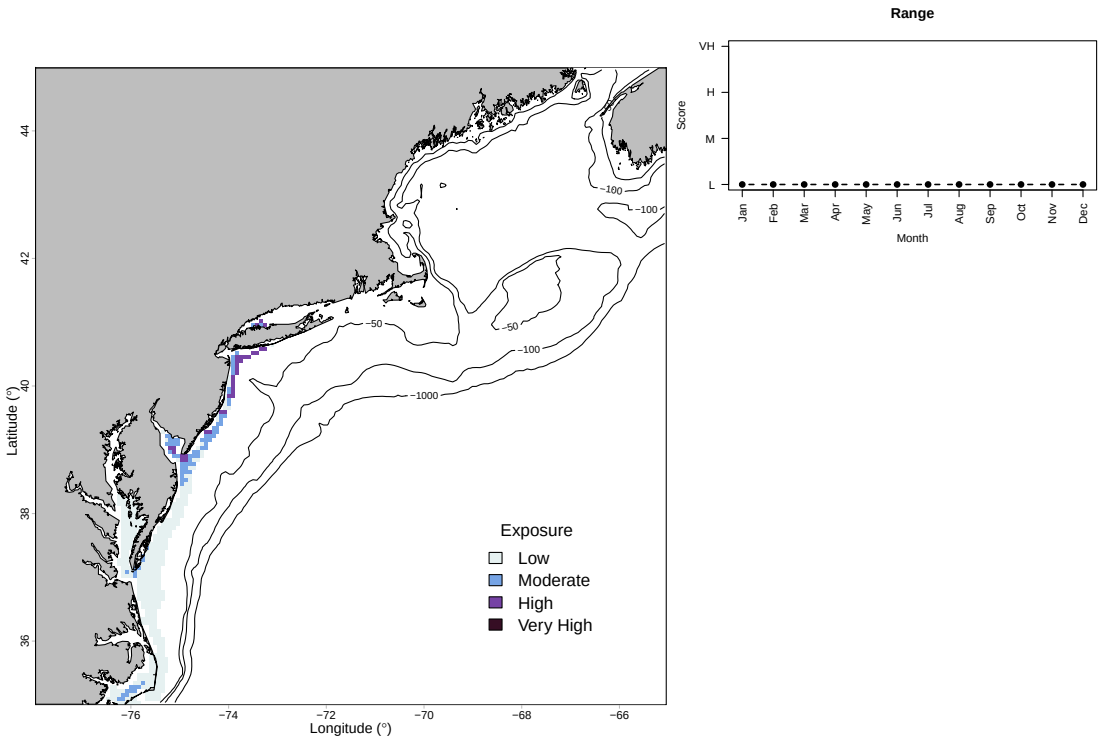


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Blue Crab distributions

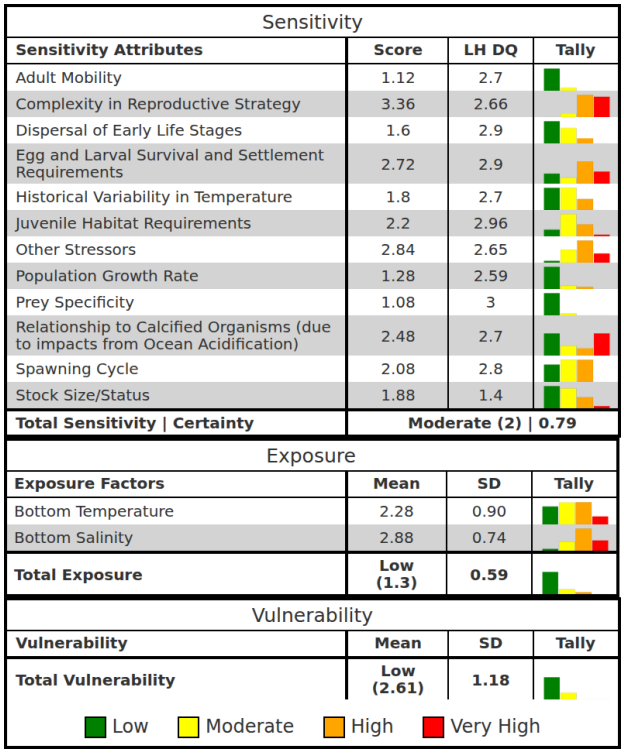


Figure 11: As in Figure 1 for Blue Crab, but with only significant Exposure factors and the resulting Vulnerability Scores.

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